

## Solution Requirements

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 5 July 2026                                                 |
| <b>Team ID</b>       | xxxxxx                                                      |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 4 Marks                                                     |

### Define Functional and Non-Functional Requirements:

#### Functional Requirements Statement

Functional requirements define the features and functions that the OptiCrop Smart Agricultural Production Optimization System must perform, such as user authentication, soil data processing, crop prediction, and report generation.

#### Non-Functional Requirements Statement

Non-functional requirements define the quality attributes of the system, such as performance, security, reliability, usability, scalability, and availability, ensuring the system operates efficiently and provides a good user experience.

#### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                                                                                | Priority (High/Medium/Low) |
|------|----------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| 1    | Authentication       | Users can securely register and log in to access the system.                                           | High                       |
| 2    | Authorization levels | Admin manages datasets and models, while farmers can enter soil details and view crop recommendations. | High                       |

|          |                                   |                                                                                            |        |
|----------|-----------------------------------|--------------------------------------------------------------------------------------------|--------|
| <b>3</b> | External interfaces               | The system interacts with the machine learning model and agricultural dataset.             | Medium |
| <b>4</b> | Transactions processing           | The system processes user inputs and generates crop recommend                              | High   |
| <b>5</b> | Reporting                         | Generate crop prediction reports with confidence score and recommendations.                | High   |
| <b>6</b> | Business rules, etc.              | Crop recommendations are based on soil nutrients, pH, temperature, humidity, and rainfall. | High   |
| <b>7</b> | Compliance to laws or regulations | Protect user data and follow data privacy guidelines.                                      | Medium |
| <b>8</b> | <i>Other</i>                      | Allow future integration of weather APIs and smart farming features.                       | Low    |

## Step 2: Non-Functional Requirements (NFR)

| S.No | NFR Category               | Requirement Description                          | Target Metric / Acceptance Criteria      |
|------|----------------------------|--------------------------------------------------|------------------------------------------|
| 1    | Performance & Speed        | Generate crop recommendations quickly.           | Response time less than <b>2 seconds</b> |
| 2    | Scalability                | Support multiple users simultaneously.           | Supports <b>1000+ users</b>              |
| 3    | Security & Data Privacy    | Protect user information and prediction data.    | Secure login and encrypted data          |
| 4    | Reliability & Availability | Ensure the application is available when needed. | <b>99% uptime</b>                        |
| 5    | Usability & Accessibility  | Provide a simple and user-friendly interface.    | Easy to use on desktop                   |

|   |              |                                                 |                                       |
|---|--------------|-------------------------------------------------|---------------------------------------|
| 6 | <i>Other</i> | Ensure maintainability and future enhancements. | Easy to update ML models and datasets |
|---|--------------|-------------------------------------------------|---------------------------------------|